Altair Vehicle Design Executive Summary

Altair Project Office
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Introduction

The Altair Lunar Lander design was created through a concerted effort by a multi-disciplinary team spread across the agency. The team pursued an approach that started with designing a minimally functional vehicle (bare necessities required to deliver people and cargo to the Moon and return them to Earth), followed by incremental functional improvements to improve its risk posture and functionality. At the time of writing this report, the Altair Project Office has completed four design cycles and two requirement analysis cycles, which spanned a duration of over three years. This executive summary gives an overview of the final state of the vehicle's functionality and design at the end of Design Analysis Cycle 4 (DAC-4).

Functional Overview

The Altair Lunar Lander's primary architectural role in crewed missions is to contain a payload during launch from earth, dock to CEV in Low Earth Orbit (LEO), insert itself and the Crew Exploration Vehicle (CEV) into Low Lunar Orbit (LLO), deliver the crew and payload to the lunar surface, sustain the crew while on the surface (in a Sortie mission mode), allow for Extra-Vehicular Activities (EVAs), and return the crew and payload from the surface to the CEV in LLO at the end of the surface mission. Alternatively, in an uncrewed mission mode, the Altair's primary architectural role is to house a payload during launch from earth, insert itself and its payload into LLO, deliver the payload to the surface, and keep itself and the payload conditioned until lunar surface assets can assume the responsibility to support the payload. The basic requirements set implied from this functional description set the basic parameter list that resulted in the initial rough calculations of the size and mass of the Altair Lunar Lander.

Ultimately, the Altair Lunar Lander was designed to satisfy hundreds of requirements. However, Table 1 provides a good summary of the vehicle's primary functionality:

LNDR.1: Provide Transportation	LNDR.1.1 : Deliver Cargo - Unpressurized
	LNDR.1.2 : Deliver Flight Crew Equipment/Pressurized Cargo
	LNDR.1.3 : Transport Crew
	LNDR.1.4 : Return Cargo
	LNDR.1.5 : Transport EVA Suits
	LNDR.1.6 : Transport Orion into Lunar Destination Orbit
LNDR.2: Support Habitability	LNDR.2.1 : Provide Exterior Viewing
	LNDR.2.2 : Control Lighting
	LNDR.2.4 : Provide Waste Management
	LNDR.2.5 : Provide Galley Service
LNDR.3: Perform Maneuvers	LNDR.3.1 : Perform Navigation and Attitude Determination
	LNDR.3.2 : Perform Translational Maneuvers
	LNDR.3.3 : Perform Attitude Control
	LNDR.3.4 : Perform Guidance

LNDR.4: Sustain Environment	LNDR.4.1 : Control Pressure
	LNDR.4.2 : Control Temperature
	LNDR.4.3 : Manage O2 and N2
	LNDR.4.4 : Control Humidity
	LNDR.4.5 : Provide Air Circulation
	LNDR.4.6 : Control Airborne Contaminants
	LNDR.4.7 : Monitor Air Quality
	LNDR.4.8 : Provide Fire Detection and Supression
	LNDR.4.9 : Provide Potable Water
LNDR.5: Accept and Execute Commands	
LNDR.6: Manage Vehicle Performance	LNDR.6.1 : Generate Health and Status
	LNDR.6.2 : Perform Fault Detection, Isolation and Recovery
	LNDR.6.3 : Perform Abort Monitoring and Execution
	LNDR.6.4: Manage Vehicle Systems
	LNDR.6.5 : Provide Health/Status Information
	LNDR.6.6 : Record Altair Data
LNDR.7: Perform Communications	LNDR.7.1 : Communicate with Constellation Systems
	LNDR.7.2 : Measure Range and Range Rate
	LNDR.7.3 : Provide Audio Services
	LNDR.7.4 : Provide Motion Imagery

Table 1. Functional Summary of the Altair Lunar Lander.

The full set of requirements levied by the Constellation Program Office and the Altair Project Office can be found in its entirety in Volume 2 of the Altair Project Report.

The requirements list represents one end product of an effort to describe how the Altair vehicle needed to function. Other end products include the operational models and modes matrices that can also be found in Volume 2. However, it is also important to understand how the Altair team arrived at this requirement set. Unlike most projects, the Altair Project Office started with basic functional requirements and went straight into a design activity to develop a basic vehicle concept, given a minimally functional mindset. From that point, the team added improvements to the vehicle to reduce the probability of Loss of Crew (LOC), followed by improvements to reduce the probability of Loss of Mission (LOM), followed by functional improvements that were deemed of great enough benefit to the mission/crew relative to the design impacts they incurred. It was only during the last of these steps, that the Altair team went back to the Constellation requirement set to determine whether the Program had requested functionality that had not been determined as a necessary component to enable the mission. Each of these requirements were scrutinized for functional gain versus design impact. At the time of writing this report, many functions required by the program office have yet to be substantiated in terms of necessity before incorporating them into the vehicle design.

Design Overview

The Altair Lunar Lander is comprised of four distinct systems or modules: Ascent Module (AM), Airlock (AL), Descent Module (DM), and the Earth Departure Stage Adapter (EDSA). These four pieces of hardware are used in varying ways to accommodate four different Design Reference Missions (DRMs), which will be discussed below.

It is particularly important to understand that the design was matured through two separate processes. The first methodology was a "bottoms-up" design effort. As it sounds, this methodology required the design team to develop details, including a master equipment list, schematics, power timelines, etc. from a clean start. The vast majority of the Altair Project Office's resources were spent on the development of the bottoms-up design. The second methodology was through the use of a parametric tool to create a parametric answer. These estimates were generated using parametric sizing relationships from a variety of sources to generate a vehicle estimate. An important distinction between these two estimates is that the bottoms-up vehicle was developed using the minimally functional design approach, whereas the parametric estimate was always focused on the final vehicle configuration. Therefore, in the tables and graphs below, there will be an expected difference between the two types of estimates.

The following sections will provide a functional and design overview for each module.

Altair Ascent Module

The Altair AM provides a habitable environment for the crew during periods of crewed operations and performs the ascent maneuvers required to deliver the crew and payload from the lunar surface to the CEV in LLO. In DRMs when the AL is not part of the vehicle configuration, the AM also provides egress and ingress to/from the lunar surface. The AM is constructed of a composite structural shell and has entrance/exit hatches in its zenith dome and through its aft sidewall. It provides the basic life support and habitability functions to support the crew. It has the systems required for both manual and automated piloting during descent and ascent. Its propulsion system uses a hypergolic mixture of Monomethylhydrazine and Nitrogen Tetroxide (MMH/NTO) run through a modified CEV engine. The AM contains four MMH/NTO Reaction Control System (RCS) pods, each containing five jets. The Altair AM's heat rejection system is a mixture of passive (passive radiators) and active (sublimator cooled loop with a heat exchanger to the DM thermal loop) components. Figure 1 shows the DAC-4 version of the AM.

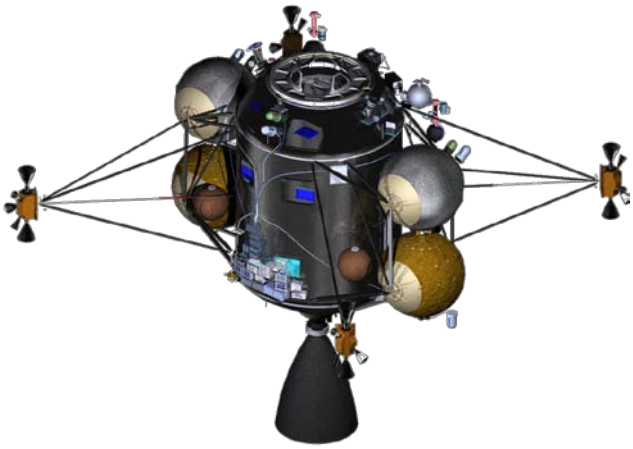


Figure 1. Altair DAC- 4 Ascent Module

Altair Airlock

The Altair AL provides both a habitable environment while the crew is inside the vehicle as well as a means to egress and ingress to/from the lunar surface. The AL is constructed of a composite structural shell and has entrance/exit hatches in its forward and port sidewalls. For the most part, it utilizes the life support systems from the AM, but provides its own basic air circulation and air monitoring functions. The AL contains a high pressure oxygen system (compressor, tanks, sensors, etc.) to provide the EVA Personal Life Support Systems (PLSSs) with 3000 psi oxygen. The AL is sized to accommodate two person donning and doffing activities. Figure 2 shows the DAC-4 version of the AL.

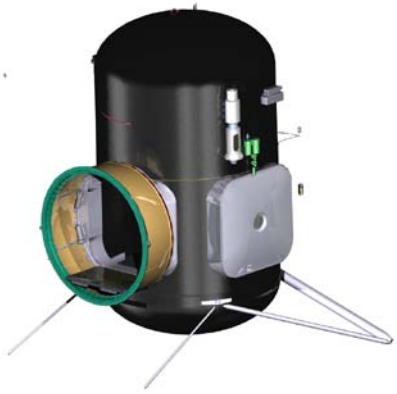


Figure 2. Altair DAC- 4 Airlock

Altair Descent Module

The Altair DM performs all the major maneuvers required to deliver the crew and payload to the lunar surface. This includes trans-lunar correction maneuvers, the lunar orbit insertion maneuver, attitude/altitude maintenance before descent, the de-orbit insertion maneuver, and powered descent through landing. The DM structure is a composite strut based structure. The landing radar is attached to the DM structure and several other sensors and avionics equipment are included depending on whether the DRM includes the Altair AM as part of the configuration. The landing gear are aluminum tubular structures with aluminum honeycome cylinders as shock absorbers in the primary struts. There are eight main propellant tanks (four liquid oxygen and four liquid hydrogen) arranged in a circular pattern. The main engine is an adjustable throttle RL-10 derivative engine. The DM contains four MMH/NTO RCS pods, each containing four jets. The Altair DM's thermal control system is a mixture of passive (passive radiators) and active (radiator cooled loop) components. The active radiator system is able to be articulated so that it can be rotated upwards towards open space in order to decrease the view factor of the surface, in the event that the thermal environment requires extra performance out of the spacecraft. Figure 3 shows the DAC-4 version of the DM.



Figure 3. Altair DAC- 4 Descent Module.

Altair Earth Departure Stage Adapter

The Altair EDSA attaches the Altair vehicle via the DM to the EDS (the upper stage and trans-lunar injection stage of the Ares V vehicle). The EDSA is a composite strut structure. The EDSA also serves as a pass-through of power and communications between the EDS and the Altair vehicle. Figure 4 shows the DAC-4 version of the EDSA.



Figure 4. Altair DAC- 4 Earth Departure Stage Adapter.

The following sections will give brief descriptions of how these modules are used to complete the four DRMs and provide mass summaries for each configuration.

Sortie DRM

The Sortie DRM requires the Altair vehicle to deliver a crew of four to the lunar surface, support the crew on the surface for a period of up to seven days, without a dependency on external systems. To enable this, the Altair vehicle uses all four of its major components: AM, AL, DM, and EDSA. Figures 5 and 6 show the DAC-4 version of the Sortie mission vehicle configuration.



Figure 5. Altair DAC-4 Sortie Mission Vehicle Configuration.

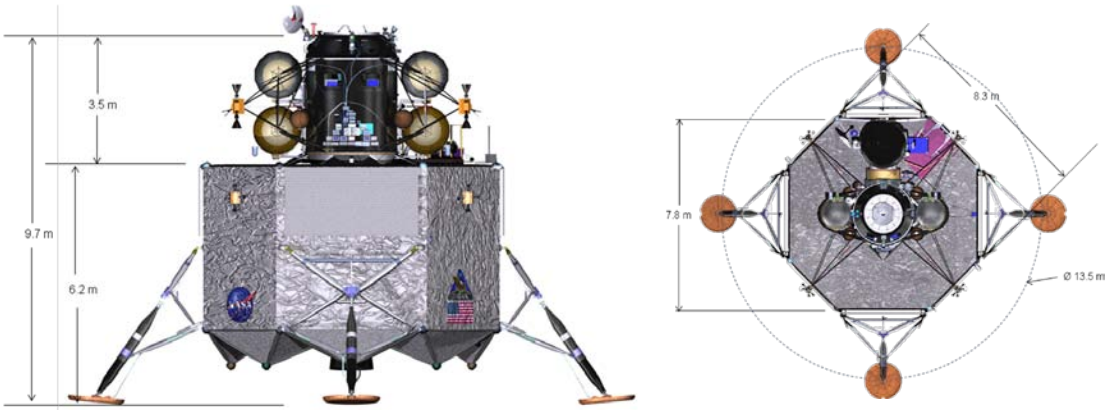


Figure 6. Dimensioned Altair DAC- 4 Sortie Mission Vehicle Configuration.

In this DRM, the EDSA separates from the DM in LEO. The DM, AM, and AL continue on their way to the moon, with the CEV attached. The DM performs the maneuvers to insert the stack into LLO, where the Altair and CEV separate. The Altair performs the maneuvers to descend to the lunar surface with the crew and payload. While on the surface, the crew lives in the AM and the AL, using the AL for access to the lunar surface. At the end of the surface mission, the AM separates from the DM and Airlock, carrying the crew and payload to orbit where it docks with the CEV. The Sortie vehicle was design to operate globally on the lunar surface.

At the conclusion of DAC-4, the Total Expected Vehicle Mass (TEVM) at the time of earth launch was 48,737 kg for the bottoms-up vehicle and 49,371 kg for the parametric vehicle, inclusive of 40% total margin (to cover mass growth allowance and manager's reserve) and enough mass to provide 90% confidence in addressing the threats/opportunities against the vehicle.

Outpost DRM

The Outpost DRM requires the Altair vehicle to deliver a crew of four to the lunar surface, stay in a quiescent state on the lunar surface for up to 210 days while the crew lives out of a surface habitat, then return the crew and payload to the CEV at the end of the surface mission. During

the surface loiter, the Altair vehicle is dependent on an external surface-based power system. The Outpost vehicle was designed to operate in the lunar polar regions. To enable this mission the Altair vehicle uses the AM, DM, and EDSA. Figures 7 and 8 show the DAC-4 version of the Outpost mission vehicle configuration.

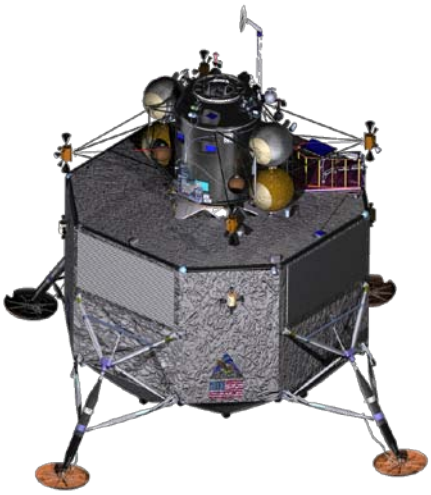


Figure 7. Altair DAC-4 Outpost Mission Vehicle Configuration.

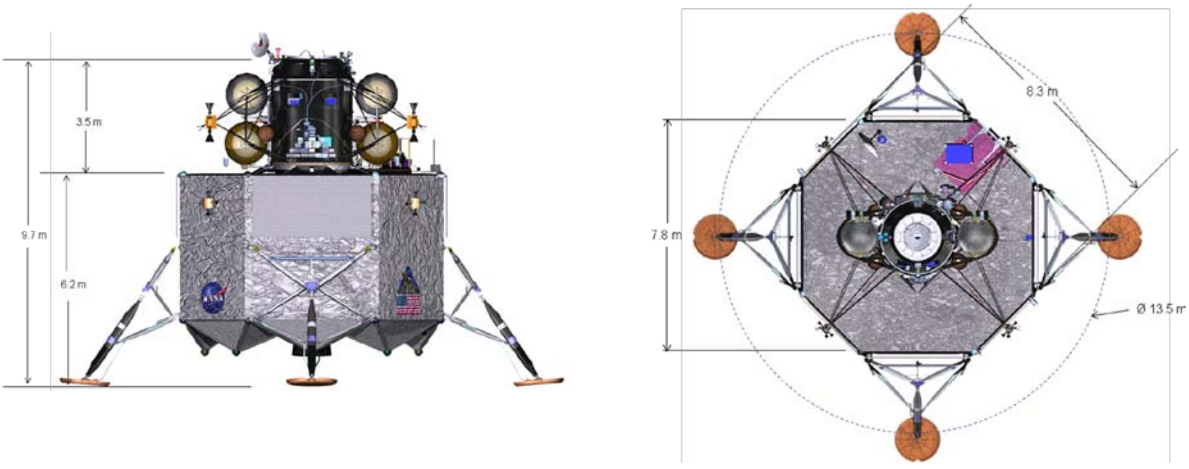


Figure 8. Dimensioned Altair DAC- 4 Outpost Mission Vehicle Configuration.

In this DRM, the EDSA separates from the DM in LEO. The DM and AM continue on their way to the moon, with the CEV attached. The DM performs the maneuvers to insert the stack into LLO, where the Altair and CEV separate. The Altair performs the maneuvers to descend to the lunar surface with the crew and payload. Once on the surface, the crew performs egress through the AM hatch and traverses to an external surface habitat after safing the Altair vehicle and establishing the power connection to the surface-based power system. At the end of the surface mission, the AM separates from the DM, carrying the crew and payload to orbit where it docks with the CEV. The Outpost vehicle was design to operate at the polar regions of the Moon.

At the conclusion of DAC-4, the TEVM at the time of earth launch was 47,323 kg for the bottoms-up vehicle, inclusive of 40% total margin (to cover mass growth allowance and manager's reserve) and enough mass to provide 90% confidence in addressing the threats/opportunities against the vehicle. The mass of the parametric estimate without protecting for threats and opportunities (a threat/opportunity assessment for this vehicle was not performed for this parametric vehicle in DAC-4) was 43,385 kg. This compares favorably with the bottoms-up version of this vehicle whose mass estimate was 43,737 kg, before threats and opportunities.

Cargo DRM

The Cargo DRM requires the Altair vehicle to deliver a payload to the lunar surface, support it for approximately three hours, and then put itself into a safe decommissioned state. To enable this mission, the Altair vehicle uses the DM and EDSA. Figures 11 and 12 show the Cargo mission vehicle configuration.



Figure 9. Altair Design Analysis Cycle 4 Cargo Mission Vehicle Configuration.

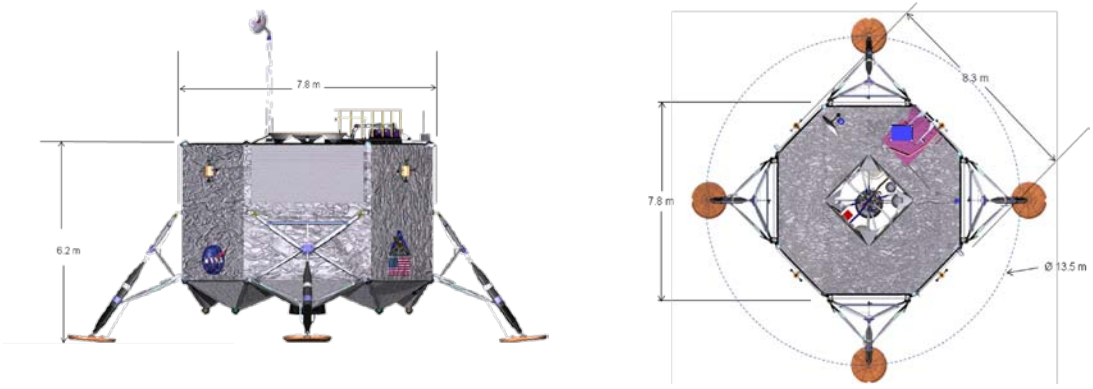


Figure 10. Dimensioned Altair Design Analysis Cycle 4 Cargo Mission Vehicle Configuration.

In this DRM, the EDSA separates from the DM in LEO. The DM continues on its way to the moon and performs descent to the lunar surface with its payload. Once on the surface, the DM continues to support the payload for approximately three hours until surface assets take over. After surface assets (e.g. surface power source) assume responsibility for the payload, the DM safes its systems and powers down.

At the conclusion of DAC-4, the total vehicle expected mass required at the time of earth launch was 55,005 kg for the parametric version of the vehicle, inclusive of 40% total margin (to cover mass growth allowance and manager's reserve) and a protection against 90% confidence in threats/opportunities against the vehicle. A bottoms-up version of this vehicle was not developed, since the parametric model was able to be used to predict an approximate cargo vehicle mass based upon Sortie data.

Extended Stay DRM

The Extended Stay DRM requires the Altair vehicle to deliver a crew of four to the lunar surface, support them for approximately three hours, stay in a quiescent state on the lunar surface for a minimum of seven days (at the time of writing this report, the maximum duration of the Extended Stay mission was under debate), and then perform ascent with the crew and payload. During the surface loiter, the Altair vehicle is dependent on an external surface-based power system. The Extended Stay vehicle was designed to operate anywhere on the lunar surface. To enable this mission, the Altair vehicle uses the AM, DM, and EDSA. Figures 9 and 10 show the Extended Stay mission vehicle configuration.

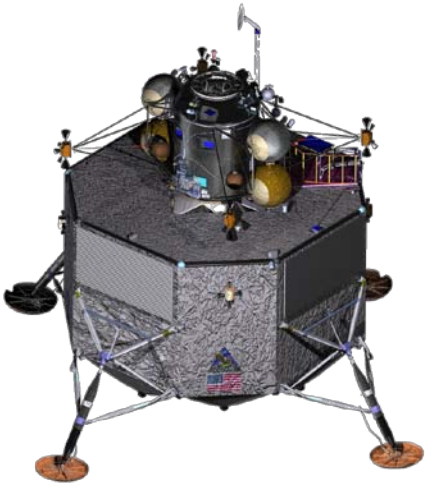


Figure 11. Altair DAC-4 Extended Stay Mission Vehicle Configuration.

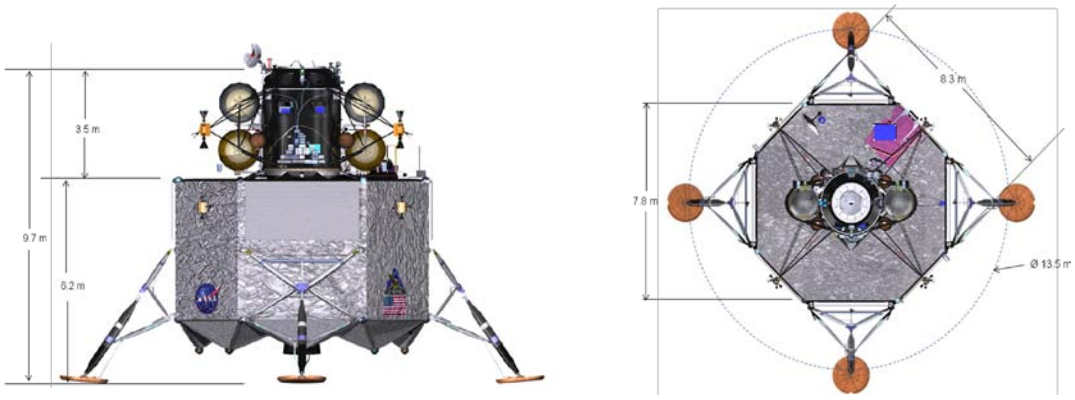


Figure 12. Dimensioned Altair DAC- 4 Extended Stay Mission Vehicle Configuration.

In this DRM, the EDSA separates from the DM in LEO. The DM and AM continue on their way to the moon, with the CEV attached. The DM performs the maneuvers to insert the stack into LLO, where the Altair and CEV separate. The Altair performs the maneuvers to descend to the lunar surface with the crew and payload. Once on the surface, the crew performs egress through the AM hatch and traverses to an external surface habitat after safing the Altair vehicle and establishing the power connection to the surface-based power system. At the end of the surface mission, the AM separates from the DM, carrying the crew and payload to orbit where it docks with the CEV. The Extended Stay vehicle was design to operate at globally on the lunar surface.

At the conclusion of DAC-4, the TEVM at the time of earth launch was 43,454 kg for the parametric version of the vehicle, inclusive of 40% total margin (to cover mass growth allowance and manager's reserve). Threats and opportunities for this parametric vehicle were not assessed. A bottoms-up version of this vehicle was not developed due to time constraints and the late development of this DRM within the Program.

Mass Summary

As the bottoms-up vehicle designs matured, its mass converged on the mass estimates of the fully functional parametric vehicle. This was expected as a result of the minimally functional buy-back process. However, with both versions of vehicle mass estimates, there is still a fair amount of maturation that would need to occur, particularly in the list of threats and opportunities logged against the designs. Note, that the mass summary data presented in Figure 13 is only for the Sortie vehicle. As a result of early design cycles, the team found that the Sortie vehicle had the least amount of mass margin relative to its control mass given the capabilities of the launch vehicle. Therefore, it became the vehicle of primary focus during each design cycle.

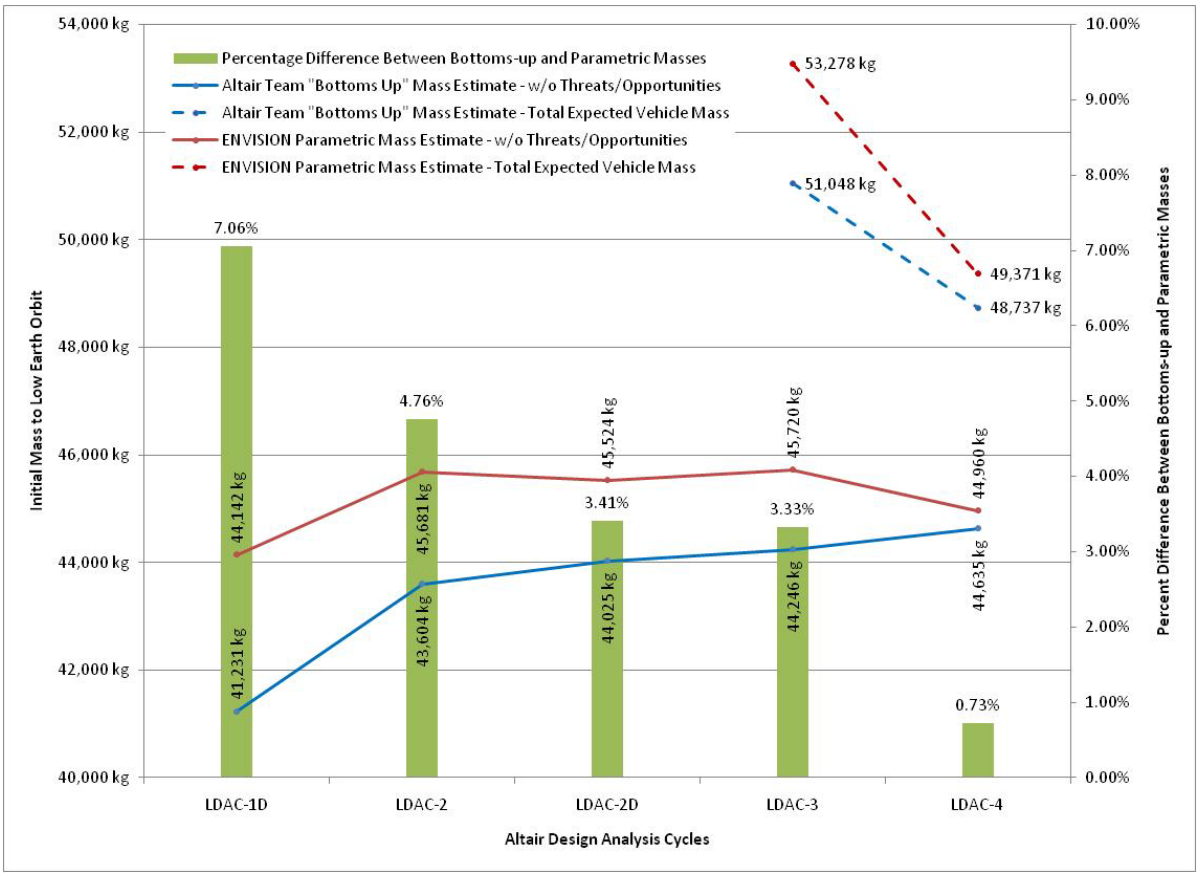


Figure 13. Sortie DRM Altair Mass History.

All of the available DRM data are summarized in Tables 2 and 3 for the bottoms-up and parametric versions of the vehicles.

Bottoms-up	Sortie DRM (kg)	Outpost DRM (kg)	Cargo DRM (kg)	Extended Stay DRM (kg)
Ascent Module	6291	6842	Not Assessed	Not Assessed
Descent Module	34069	34069		
Airlock	1358	N/A		
Earth Departure Stage Adapter	436	436		
Project Manager's Reserve	1980	1890		
Payload	500	500		
Net Threats/Opportunities (90% Confidence)	4102	3586		
Total Expected Vehicle Mass	48737	47323		

Table 2. DAC-4 Bottoms-up Total Expected Vehicle Masses For Each Design Reference Mission.

Parametric	Sortie DRM (kg)	Outpost DRM (kg)	Cargo DRM (kg)	Extended Stay DRM (kg)
Ascent Module	6352	7281	N/A	7281
Descent Module	34386	32505	36267	33664
Airlock	1362	0	N/A	0
Earth Departure Stage	488	488	488	488

Adapter				
Project Manager's Reserve	1872	1754	1382	1754
Payload	500	500	14500	500
Net Threats/Opportunities (90% Confidence)	4412	3559	2856	3599
Total Expected Vehicle Mass	49372	46087	55493	47286

Table 3. DAC-4 Parametric Total Expected Vehicle Masses for Each Design Reference Mission.

Conclusion

After three years of study, the Altair Project Office has developed a firm grasp of the major requirements that govern the functionality of the Altair Lunar Lander and a set of vehicle sub-system designs that could perform the mission. The masses presented in this report represent the most concerted effort of the agency to develop human lander estimates since Apollo. After significant maturation of the baseline bottoms-up design, including upgrades to improve the LOC and LOM risk posture of the vehicle, the vehicle mass estimates from Altair's two different approaches began to converge, providing confidence that the mass estimates are realistic.